11. Number of Possible Playfair Keys

CODE:

```
import math

total_keys = math.factorial(25)

print("Total possible keys:", total_keys)
print("Approximate power of 2:", 2 ** round(math.log2(total_keys)))

# Equivalent keys from cyclic row and column shifts
unique_keys = math.factorial(25) // 25

print("Effectively unique keys:", unique_keys)
print("Approximate power of 2:", 2 ** round(math.log2(unique_keys)))
```

OUTPUT:

